Smith normal form over PIDs

$$\begin{tikzcd}
R^{n'} \arrow[r, phantom, "\cong"] & N' \arrow[r, phantom, "\leq"] & N \arrow[d] & \\
 & & N/N' \arrow[r, phantom, "\cong"] & R \\
 & & \boxed{n_{n'+1} + N'} & 1 \arrow[l, mapsto]
\end{tikzcd}$$

$$e_i \mapsto n_i$$

Lemma: In this situation,

$$N \cong R^{n'+1}$$

Pf: We need

$$R^{n'+1} \xrightarrow{\cong} N$$

$$e_i \longmapsto n_i \qquad 1 \leq i \leq n'$$

$$e_{n'+1} \longmapsto n_{n'+1}$$

This gives the desired isomorphism

$$\begin{tikzcd}
e_i \arrow[r, mapsto] & n_i \\
R^{n'} \times \{0\} \arrow[r, "\cong"] \arrow[d, phantom, "\rotatebox{90}{$\leq$}"] & N' \arrow[d] \\
R^{n'+1} \arrow[r, "\cong"] \arrow[d] & N \arrow[d] \\
R \arrow[r, "\cong"] & N/N' \\
1 \arrow[r, mapsto] & n_{n'+1} + N'
\end{tikzcd}$$

isomorphism

We've completed the proof of

Proposition 2 $\quad m \geq 1$

$N \leq R^m$: R-submodule.

Then $\exists\, n \leq m$ & an isomorphism of R-modules

$$R^n \xrightarrow{\cong} N$$

State of things

N : f.g. R-module

Then $\exists\, M \leq R^m$ s.t. $R^m/M \simeq N$

$$\begin{tikzcd} M \arrow[r, phantom, "\leq"] & R^m \\ R^n \arrow[u, "\simeq"] \arrow[ur, "f"'] & \end{tikzcd}$$

$$R^m/M \cong R^m/\operatorname{im}(f) = R^m/\operatorname{im}(A_f)$$

$$f \longleftrightarrow A_f = \Big[\underbrace{f(e_1) \ \cdots \ f(e_n)}_{\text{view these as } m\times 1\text{-column vectors over } R}\Big] \in M_{m\times n}(R)$$

<u>Observation:</u> $B \in GL_n(R)$, $C \in GL_m(R)$

$$GL_n(R) = \left\{ A \in M_{n\times n}(R) : \begin{array}{l} \exists\, A^{-1} \in M_{n\times n}(R) \\ \text{s.t.} \\ A^{-1}A = AA^{-1} = I_n \end{array} \right\}$$

Then $R^m/\operatorname{im}(A_f) \simeq R^m/\operatorname{im}(CA_fB)$

Proposition: We can find C, B s.t.

$$CA_f B = \begin{bmatrix} r_1 & & & & 0 & 0 \\ & r_2 & & & 0 & 0 \\ & & \ddots & & & \\ 0 & & & r_t & & \\ & & & & 0 & \\ 0 & 0 & & & & \ddots & 0 \end{bmatrix}$$

where $r_1 | r_2 | \dots | r_t$

Rmk: Multiplying by C on left $\longleftrightarrow$ Row operations

Multiplying by B on right $\longleftrightarrow$ Column operations

Row operations

(i) Add r-times the j-th row to the i-th row

$(i \neq j \ , \ r \in R)$

(ii) Switch the i-th row with j-th row

(iii) $A = \begin{bmatrix} a & \cdots & \cdots \\ b & \cdots & \cdot \\ \vdots & & \\ \vdots & \cdot & \cdot \end{bmatrix}$

$d = \gcd(a,b) = sa + tb, \quad \exists s,t \in R$

$\Downarrow$

$1 = s\left(\frac{a}{d}\right) + t\left(\frac{b}{d}\right)$

$\Rightarrow \begin{bmatrix} s & t \\ -\frac{b}{d} & \frac{a}{d} \end{bmatrix} \in GL_2(R)$

$$\begin{bmatrix} s & t \\ -b/d & a/d \end{bmatrix}^{-1} = \begin{bmatrix} a/d & -t \\ b/d & s \end{bmatrix}$$

Rmk: the meaning of "$\frac{a}{d}$" is the element s.t. $\frac{a}{d} \cdot d = a$.

$$\begin{bmatrix} s & t & 0 \\ -b/d & a/d & \\ 0 & & I \end{bmatrix} \begin{bmatrix} a & \cdots & \cdots \\ b & & \\ \vdots & & \end{bmatrix} = \begin{bmatrix} d & e & \cdots \\ 0 & \sim & \\ \vdots & & \sim \end{bmatrix}$$

Rmk (GCD in PIDs)

$r, s \in R$; we can consider

$$(r) + (s) = (t)$$

where $(r) \leq (t) \iff r = t\underset{\in R}{a}$

$(s) \leq (t) \iff s = tb \in R$.

But also

$$t = q_1 r + q_2 s \quad \exists q_1, q_2 \in R.$$
$$\in (r) + (s).$$

$\Rightarrow$ If u is s.t. $u \mid r$ & $u \mid s$

then $u \mid (q_1 r + q_2 s) = t$.

$\boxed{t = \gcd(r, s)}$ [Only well-defined up to invertible elements]

Pf of proposition

(i) A = zero matrix

(ii) By switching rows & columns, we set

$$A = \begin{bmatrix} \overset{\neq 0}{\boxed{a_0}} & e & \cdots \\ \sim & & \sim \\ \sim & & \end{bmatrix}$$

(iii) If a_0 divides all the other entries in the first row and column, we leave it be.

(iv) Otherwise, we can use operations (ii) & (iii) to replace a_0 with the gcd of all the entries in the first row & first column

(v) Now we can use operation (i) to make sure we have

$$A = \begin{bmatrix} a_0 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & * & \\ 0 & & & \end{bmatrix}$$

(vi) By induction, we can put A in the form

$$\begin{bmatrix} c_0 & & & & & 0 \\ & r_2 & & & & \\ & & \ddots & & & \\ & & & \ddots & & \\ & 0 & & & r_t & \\ & & & & & \ddots \; 0 \end{bmatrix}$$

$r_2 \mid r_3 \mid \cdots \mid r_t$.

(vii) We now add the other rows to the first row & repeat previous steps to replace c_0 with the gcd of all the entries of the matrix.

(viii) Repeat step (vi) again.